

SF光机投影时序计算说明

公式：

- 。 The listed minimum dark times are applicable for the minimum exposure time of each sequence. The dark times scale along with the exposure time as per the following equation:

- $$T_{\text{min required pre-exposure dark time}} = t_{\text{sequence min pre-exposure dark time}} * (T_{\text{desired exposure}} / t_{\text{sequence minimum exposure}}) + 1 \text{ (to round up if needed)}$$
- $$T_{\text{min required post-exposure dark time}} = t_{\text{sequence min post-exposure dark time}} * (T_{\text{desired exposure}} / t_{\text{sequence minimum exposure}}) + 1 \text{ (to round up if needed)}$$

- 举例：我的图片类型是8位单色图，设置的曝光时间为3000us
- 根据右边图片可行，SEQUENCE INDEX为22
- $T_{\text{desired exposure}} = 3000$
- $t_{\text{sequence minimum exposure}} = 2084$ （INDEX22区间的最小值）
- $t_{\text{sequence min pre-exposure dark time}} = 171$
- $t_{\text{sequence min post-exposure dark time}} = 31$

Pattern Type	Sequence Index	Exposure Time Range (us)	Sequence Min Pre-Exposure Dark Time (us)	Sequence Min Post-Exposure Dark Time (us)
1-bit Monochrome	21	200 - 399	171	31
	20	400 - 799	171	31
	19	800 - 1799	171	31
	18	1800 - 3799	171	31
	17	3800 - 7799	171	31
	16	7800 - 15799	171	31
	15	15800 - 31799	171	31
	14	31800 - 63799	171	31
	13	63800 - 127600	171	31
1-bit RGB	27	600 - 1799	171	31
	26	1800 - 3799	171	31
	25	3800 - 7600	171	31
8-bit Monochrome	22	2084 - 4168	171	31
	23*	1677 - 2083	171	31
8-bit RGB	24	6512 - 13024	171	31

- $T_{\text{min required pre-exposure dark time}} = t_{\text{sequence min pre-exposure dark time}} * (T_{\text{desired exposure}} / t_{\text{sequence minimum exposure}}) + 1$ (to round up if needed)
- $T_{\text{min required post-exposure dark time}} = t_{\text{sequence min post-exposure dark time}} * (T_{\text{desired exposure}} / t_{\text{sequence minimum exposure}}) + 1$ (to round up if needed)

- 将“Tdesired exposure =3000
- tsequence minimum exposure=2084 (INDEX22区间的最小值)
- tsequence min pre-exposure dark time =171
- tsequence min post-exposure dark time =31“
- 代入以上公式得：

- $T_{\text{min required pre-exposure dark time}} = 171 * (3000/2084) + 1 = 247$
- $T_{\text{min required post-exposure dark time}} = 31 * (3000/2084) + 1 = 45$